VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum- 590014, Karnataka.

LAB RECORD

on

Big Data Analytics (23CS6PCBDA)

Submitted by

Shreya Raj (1BM23CS317)

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
BENGALURU - 560019
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B.M.S. College of Engineering

Bull Temple Road, Bangalore 560019
(Affiliated to Visvesvaraya Technological University, Belgaum)

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Big Data Analytics" carried out by **Shreya Raj (1BM23CS317),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2026. The Lab report has been approved as it satisfies the academic requirements in respect of a Big Data Analytics (23CS6PCBDA) work prescribed for the said degree.

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INDEX

Sl. No.	Date	Experiment Title	Page No.
1	04.03.25	MongoDB- CRUD Operations Demonstration (Practice and Self Study)	1

2	01.04.25	Perform the following DB operations using Cassandra.
		• Create a keyspace by name Employee
		• Create a column family by name
		  ○ Employee-Info with attributes
		  ○ Emp_Id Primary Key, Emp_Name, Designation,
		  ○ Date_of_Joining, Salary, Dept_Name
		• Insert the values into the table in batch
		• Update Employee name and Department of EmpId 121
		• Sort the details of Employee records based on salary
		• Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
		• Update the altered table to add project names.
		• Create a TTL of 15 seconds to display the values of Employees.
			7

3	08.04.25	Perform the following DB operations using Cassandra.
		• Create a keyspace by name Library
		• Create a column family by name Library-Info with attributes
		  ○ Stud_Id Primary Key,
		  ○ Counter_value of type Counter,
		  ○ Stud_Name, Book-Name, Book-Id,
		  ○ Date_of_issue
		• Insert the values into the table in batch
		• Display the details of the table created and increase the value of the counter
		• Write a query to show that a student with id 112 has taken a book “BDA” 2 times.
		• Export the created column to a csv file
		• g) Import a given csv dataset from local file system into Cassandra column family
			10

4	15.04.25	Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)	17
5	15.04.25	Implement Wordcount program on Hadoop framework	20
6	06.05.25	From the following link extract the weather data https://github.com/tomwhite/hadoop-book/tree/master/input/ncdc/all • Create a MapReduce program to find average temperature for each year from NCDC data set. • b) find the mean max temperature for every month	24

7	20.05.25	For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words.	33
8	20.05.25	Write a Scala program to print numbers from 1 to 100 using for loop.	38
9	20.05.25	Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.	39
10	20.05.25	Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).	40

Github Link: https://github.com/RajShreyaa/1BM23CS317_BDA_Lab

Course Outcomes (COs):

CO1	Apply the concept of NoSQL, Hadoop or Spark for a given task
CO2	Analyse big data analytics mechanisms that can be applied to obtain solution for a given problem.
CO3	Design and implement solutions using data analytics mechanisms for a given problem.

LABORATORY PROGRAM – 1

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

COMMAND WITH OUTPUT - USING ATLAS

```
>_MONGOSH
> db["Employee_Info"].find()
<
> use Employee
< switched to db Employee
> db.createCollection("Employee_Info")
< { ok: 1 }
> db.Employee_Info.createIndex({ Emp_Id: 1 }, { unique: true })
✖ › MongoServerError[DuplicateKey]: Index build failed: 5778370
> db.Employee_Info.insertMany([
    {
      _id: 121,
      Emp_Name: "Alice",
      Designation: "Software Engineer",
      Date_of_Joining: ISODate("2021-06-15"),
      Salary: 75000,
      Dept_Name: "IT"
    },
    {
      _id: 122,
      Emp_Name: "Bob",
      Designation: "Senior Engineer",
      Date_of_Joining: ISODate("2020-03-10"),
      Salary: 90000,
      Dept_Name: "R&D"
    },
    {
      _id: 123,
      Emp_Name: "Charlie",
      Designation: "Manager",
      Date_of_Joining: ISODate("2019-01-20"),
      Salary: 105000,
      Dept_Name: "HR"
    }
  ])
```

```
< {
    acknowledged: true,
    insertedIds: {
      '0': 121,
      '1': 122,
      '2': 123
    }
  }
> db.Employee_Info.updateOne(
    { _id: 121 },
    {
      $set: {
        Emp_Name: "Alice Johnson",
        Dept_Name: "Engineering"
      }
    }
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.Employee_Info.find().sort({ Salary: 1 })
< {
    _id: 121,
    Emp_Name: 'Alice Johnson',
    Designation: 'Software Engineer',
    Date_of_Joining: 2021-06-15T00:00:00.000Z,
    Salary: 75000,
    Dept_Name: 'Engineering'
  }
  {
    _id: 122,
    Emp_Name: 'Bob',
    Designation: 'Senior Engineer',
```

```
  {
    _id: 122,
    Emp_Name: 'Bob',
    Designation: 'Senior Engineer',
    Date_of_Joining: 2020-03-10T00:00:00.000Z,
    Salary: 90000,
    Dept_Name: 'R&D'
  }
  {
    _id: 123,
    Emp_Name: 'Charlie',
    Designation: 'Manager',
    Date_of_Joining: 2019-01-20T00:00:00.000Z,
    Salary: 105000,
    Dept_Name: 'HR'
  }
> db.Employee_Info.find().sort({ Salary: -1 })
< {
    _id: 123,
    Emp_Name: 'Charlie',
    Designation: 'Manager',
    Date_of_Joining: 2019-01-20T00:00:00.000Z,
    Salary: 105000,
    Dept_Name: 'HR'
  }
  {
    _id: 122,
    Emp_Name: 'Bob',
    Designation: 'Senior Engineer',
    Date_of_Joining: 2020-03-10T00:00:00.000Z,
    Salary: 90000,
    Dept_Name: 'R&D'
  }
  {
    _id: 121,
    Emp_Name: 'Alice Johnson',
    Designation: 'Software Engineer',
```

```
{
  _id: 121,
  Emp_Name: 'Alice Johnson',
  Designation: 'Software Engineer',
  Date_of_Joining: 2021-06-15T00:00:00.000Z,
  Salary: 75000,
  Dept_Name: 'Engineering'
}
> db.Employee_Info.updateOne(
  { _id: 121 },
  {
    $addToSet: {
      Projects: { $each: ["Payroll System", "Cloud Migration"] }
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.Employee_Info.updateMany(
  {},
  {
    $set: {
      expireAt: new Date(Date.now() + 15000)
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

```
> db.Employee_Info.updateMany(
    {},
    {
      $set: {
        expireAt: new Date(Date.now() + 15000)
      }
    }
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 3,
    modifiedCount: 3,
    upsertedCount: 0
  }
> db.Employee_Info.updateMany(
    {},
    {
      $set: {
        expireAt: new Date(Date.now() + 15000)
      }
    }
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 3,
    modifiedCount: 3,
    upsertedCount: 0
  }
> db.Employee_Info.createIndex(
    { expireAt: 1 },
    { expireAfterSeconds: 0 }
  )
< expireAt_1
>
Employee>
```

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

COMMAND WITH OUTPUT - USING UBUNTU TERMINAL

```
MyDataBase> use MyDataBase
already on db MyDataBase
MyDataBase> show collections
Customers
NewStudent
Student
MyDataBase> db.Student.find();
[
  {
    _id: 1,
    studName: 'Michellejacintha',
    Grade: 'VII',
    Hobbies: 'InternetSurfing'
  },
  { _id: 3, Grade: 'VII', StudName: 'AryanDavid', Hobbies: 'Skating' },
  { _id: 2, Grade: 'VIII', StudName: 'Ram', Hobbies: 'Learning' }
]
```

```
test> use MyDataBase
switched to db MyDataBase
MyDataBase> show collections
NewStudent
NewStudent2
Student
MyDataBase> db.NewStudent2.drop();
true
MyDataBase> db.createCollection("Customers");
{ ok: 1 }
MyDataBase> db.Customers.insertMany([{cust_id:1,Balance:200, Type:"S"},]);
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d00571207666297fa3b81a') }
}
MyDataBase> db.Customers.insert({cust_id:1,Balance:1000, Type:"Z"})
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d0058f207666297fa3b81b') }
}
MyDataBase> db.Customers.insert({cust_id:2,Balance:100, Type:"Z"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d0059c207666297fa3b81c') }
}
MyDataBase> db.Customers.insert({cust_id:2,Balance:1000, Type:"C"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d005a5207666297fa3b81d') }
}
MyDataBase> db.Customers.insert({cust_id:2,Balance:500, Type:"C"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d005ad207666297fa3b81e') }
}
MyDataBase> db.Customers.insert({cust_id:2,Balance:50, Type:"S"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d005b2207666297fa3b81f') }
}
MyDataBase> db.Customers.insert({cust_id:3,Balance:500, Type:"Z"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d005ba207666297fa3b820') }
}
```

```
MyDataBase> db.Customers.aggregate([
...   {
...     $group: {
...       _id: "$cust_id",              // Group by cust_id
...       minAccBal: { $min: "$Balance" }, // Find the minimum Balance
...       maxAccBal: { $max: "$Balance" }  // Find the maximum Balance
...     }
...   }
... ]);
[
  { _id: 3, minAccBal: 500, maxAccBal: 500 },
  { _id: 2, minAccBal: 50, maxAccBal: 1000 },
  { _id: 1, minAccBal: 200, maxAccBal: 1000 }
]
```

```
MyDataBase> db.Customers.aggregate([
...   { $match: { Type: "Z" } },
...   { $group: { _id: "$cust_id", TotAccBal: { $sum: "$Balance" } } },
...   { $match: { TotAccBal: { $gt: 1200 } } }
... ]);
```

```
MyDataBase> db.Customers.aggregate([
...   { $match: { Type: "Z" } },
...   {
...     $group: {
...       _id: "$cust_id",
...       TotAccBal: { $sum: "$Balance" }
...     }
...   },
...   {
...     $match: {
...       TotAccBal: { $gt: 200 }
...     }
...   }
... ]);
[ { _id: 3, TotAccBal: 500 }, { _id: 1, TotAccBal: 1000 } ]
```

```
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ mongoimport --host localhost --db MyDataBase --coll
ection NewStudent2 --type=csv --file /home/bmscecse/Desktop/135.txt --headerline
2025-03-11T14:55:05.192+0530    connected to: mongodb://localhost/
2025-03-11T14:55:05.360+0530    3 document(s) imported successfully. 0 document(s) failed to import.
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ mongoexport --host localhost --db MyDataBase --coll
ection NewStudent2 --type=json --file /home/bmscecse/Desktop/135.txt
2025-03-11T14:55:24.438+0530    error parsing command line options: unknown option "file"
2025-03-11T14:55:24.438+0530    try 'mongoexport --help' for more information
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ mongoexport --host localhost --db MyDataBase --coll
ection NewStudent2 --type=json --out /home/bmscecse/Desktop/135.txt
2025-03-11T14:55:32.771+0530    connected to: mongodb://localhost/
2025-03-11T14:55:32.780+0530    exported 3 records
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

LABORATORY PROGRAM – 2

Perform the following DB operations using Cassandra

Questions:

a) Create a keyspace by name Employee
b) Create a column family by name
- Employee-Info with attributes
- Emp_Id Primary Key, Emp_Name,
- Designation, Date_of_Joining,
- Salary, Dept_Name

c) Insert the values into the table in batch
d) Update Employee name and Department of Emp-Id 121
e) Sort the details of Employee records based on salary
f) Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
g) Update the altered table to add project names.
h) Create a TTL of 15 seconds to display the values of Employees.

COMMAND WITH OUTPUT

```
mongosh mongodb://127.0.0.
icrosoft Windows [Version 10.0.26100.7623]
(c) Microsoft Corporation. All rights reserved.

:\Users\student>mongosh
urrent Mongosh Log ID: 6989a2879a3a13ba051e2620
onnecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.10
sing MongoDB:          8.0.11
sing Mongosh:          2.5.10
ongosh 2.6.0 is available for download: https://www.mongodb.com/try/download/shell

or mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-02-09T08:35:42.510+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

est> use Employee
witched to db Employee
mployee> db.Employee_Info.insertMany([
.. {
..   Emp_Id:121,
..   Emp_Name:"Amit",
..   Designation:"Software Engineer",
..   Date_of_Joining:new Date("2019-06-15"),
..   Salary:60000,
..   Dept_Name:"IT"
.. },
.. {
..   Emp_Id:122,
..   Emp_Name:"Ravi",
..   Designation:"Senior Engineer",
..   Date_of_Joining:new Date("2018-03-20"),
..   Salary:75000,
..   Dept_Name:"R&D"
.. },
.. {
..   Emp_Id:123,
..   Emp_Name:"Meera",
..   Designation:"Manager",
..   Date_of_Joining:new Date("2016-01-10"),
..   Salary:90000,
..   Dept_Name:"HR"
.. }
.. ])
..

 acknowledged: true,
 insertedIds: {
   '0': ObjectId('6989a2be9a3a13ba051e2621'),
   '1': ObjectId('6989a2be9a3a13ba051e2622'),
```

```
mongosh mongodb://127.0.0.
Employee> db.Employee_Info.find()
[
  {
    _id: ObjectId('6989a2be9a3a13ba051e2621'),
    Emp_Id: 121,
    Emp_Name: 'Amit',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'IT'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2622'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2623'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  }
]
Employee> db.Employee_Info.find().pretty()
[
  {
    _id: ObjectId('6989a2be9a3a13ba051e2621'),
    Emp_Id: 121,
    Emp_Name: 'Amit',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'IT'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2622'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
```

```
mongosh mongodb://127.0.0.
Employee> db.Employee_Info.find().pretty()
[
  {
    _id: ObjectId('6989a2be9a3a13ba051e2621'),
    Emp_Id: 121,
    Emp_Name: 'Amit',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'IT'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2622'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2623'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  }
]
Employee> db.Employee_Info.updateOne(
...   {Emp_Id:121},
...   {$set:{Emp_Name:"Amit Kumar", Dept_Name:"Data Science"}}
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Employee> db.Employee_Info.find({Emp_Id:121})
[
  {
    _id: ObjectId('6989a2be9a3a13ba051e2621'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
```

```
mongosh mongodb://127.0.0.
Employee> db.Employee_Info.updateOne(
..   {Emp_Id:121},
..   {$set:{Emp_Name:"Amit Kumar", Dept_Name:"Data Science"}}
.. )
..
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Employee> db.Employee_Info.find({Emp_Id:121})
[
  {
    _id: ObjectId('6989a2be9a3a13ba051e2621'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science'
  }
]
Employee> db.Employee_Info.find().sort({Salary:-1})
[
  {
    _id: ObjectId('6989a2be9a3a13ba051e2623'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2622'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2621'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
```

```
mongosh mongodb://127.0.0.
mployee> db.Employee_Info.updateOne(
..   {Emp_Id:121},
..   {$set:{
..     Projects:["AI-Analytics","Cloud-Migration","BigData-Pipeline"]
..   }}
.. )
..

 acknowledged: true,
 insertedId: null,
 matchedCount: 1,
 modifiedCount: 1,
 upsertedCount: 0

mployee> db.Employee_Info.find().sort({Salary:1})

 {
   _id: ObjectId('6989a2be9a3a13ba051e2621'),
   Emp_Id: 121,
   Emp_Name: 'Amit Kumar',
   Designation: 'Software Engineer',
   Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
   Salary: 60000,
   Dept_Name: 'Data Science',
   Projects: [ 'AI-Analytics', 'Cloud-Migration', 'BigData-Pipeline' ]
 },
 {
   _id: ObjectId('6989a2be9a3a13ba051e2622'),
   Emp_Id: 122,
   Emp_Name: 'Ravi',
   Designation: 'Senior Engineer',
   Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
   Salary: 75000,
   Dept_Name: 'R&D'
 },
 {
   _id: ObjectId('6989a2be9a3a13ba051e2623'),
   Emp_Id: 123,
   Emp_Name: 'Meera',
   Designation: 'Manager',
   Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
   Salary: 90000,
   Dept_Name: 'HR'
 }

mployee> db.Employee_Info.insertOne({
..   Emp_Id:130,
..   Emp_Name:"TemporaryUser",
..   createdAt:new Date()
.. })
..
```

```
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2623'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  }
]
Employee> db.Employee_Info.insertOne({
...   Emp_Id:130,
...   Emp_Name:"TemporaryUser",
...   createdAt:new Date()
... })
...
{
  acknowledged: true,
  insertedId: ObjectId('6989a3909a3a13ba051e2624')
}
Employee> db.Employee_Info.createIndex(
...   {createdAt:1},
...   {expireAfterSeconds:15}
... )
...
createdAt_1
Employee> db.Employee_Info.find({Emp_Id:130})
[
  {
    _id: ObjectId('6989a3909a3a13ba051e2624'),
    Emp_Id: 130,
    Emp_Name: 'TemporaryUser',
    createdAt: ISODate('2026-02-09T09:06:24.471Z')
  }
]
Employee> show dbs
Employee        108.00 KiB
Employee_Info    72.00 KiB
admin            40.00 KiB
config          108.00 KiB
e_commerce_db   120.00 KiB
local            96.00 KiB
travelmate       40.00 KiB
Employee>
```

LABORATORY PROGRAM – 3

Perform the following DB operations using Cassandra

Questions:

a) Create a keyspace by name Library
b) Create a column family by name Library-Info with attributes

- Stud_Id Primary Key,
- Counter_value of type Counter,
- Stud_Name, Book-Name, Book-Id,
- Date_of_issue

c) Insert the values into the table in batch
d) Display the details of the table created and increase the value of the counter
e) Write a query to show that a student with id 112 has taken a book "BDA" 2 times.
f) Export the created column to a csv file
g) Import a given csv dataset from local file system into Cassandra column family

COMMAND WITH OUTPUT

```
cqlsh> CREATE KEYSPACE Library
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
```

```
cqlsh> USE Library;
```

```
cqlsh:library> CREATE TABLE Library_Info (
           ...     Stud_Id int,
           ...     Stud_Name text,
           ...     Book_Name text,
           ...     Book_Id int,
           ...     Date_of_issue date,
           ...     PRIMARY KEY (Stud_Id, Book_Id)
           ... );
cqlsh:library> CREATE TABLE Library_Counter (
           ...     Stud_Id int,
           ...     Book_Name text,
           ...     Counter_value counter,
           ...     PRIMARY KEY (Stud_Id, Book_Name)
           ... );
```

```
cqlsh:library> BEGIN BATCH
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id,
 Date_of_issue)
           ... VALUES (112, 'Rahul', 'BDA', 101, '2026-02-15');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id,
 Date_of_issue)
           ... VALUES (113, 'Amit', 'DBMS', 102, '2026-02-14');
           ...
           ... APPLY BATCH;
```

```
cqlsh:library> UPDATE Library_Counter
            ... SET Counter_value = Counter_value + 1
            ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> UPDATE Library_Counter
            ... SET Counter_value = Counter_value + 1
            ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
```

```
cqlsh:library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue | stud_name
---------+---------+-----------+---------------+-----------
     113 |     102 |      DBMS |    2026-02-14 |      Amit
     112 |     101 |       BDA |    2026-02-15 |     Rahul

(2 rows)
cqlsh:library> SELECT * FROM Library_Counter;

 stud_id | book_name | counter_value
---------+-----------+---------------
     112 |       BDA |             2
```

```
cqlsh:library> SELECT Counter_value
            ... FROM Library_Counter
            ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';

 counter_value
---------------
             2
```

```
Starting copy of library.library_info with columns [stud_id, book_id, book_name,
 date_of_issue, stud_name].
Processed: 2 rows; Rate:      14 rows/s; Avg. rate:      14 rows/s
2 rows exported to 1 files in 0.169 seconds.
cqlsh:library> COPY Library_Counter TO 'library_counter.csv' WITH HEADER = TRUE;
Using 16 child processes

Starting copy of library.library_counter with columns [stud_id, book_name, count
er_value].
Processed: 1 rows; Rate:      10 rows/s; Avg. rate:      10 rows/s
1 rows exported to 1 files in 0.120 seconds.
cqlsh:library>
```

```
cqlsh:library> COPY Library_Info
           ... (stud_id, book_id, book_name, date_of_issue, stud_name)
           ... FROM 'library_info.csv'
           ... WITH HEADER = TRUE;
Using 16 child processes

Starting copy of library.library_info with columns [stud_id, book_id, book_name,
 date_of_issue, stud_name].
Processed: 2 rows; Rate:       4 rows/s; Avg. rate:       5 rows/s
2 rows imported from 1 files in 0.387 seconds (0 skipped).
cqlsh:library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue | stud_name
---------+---------+-----------+---------------+-----------
     113 |     102 |      DBMS |    2026-02-14 |      Amit
     112 |     101 |       BDA |    2026-02-15 |     Rahul

(2 rows)
```

LABORATORY PROGRAM – 4

Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)

COMMAND WITH OUTPUT

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ cd ./Desktop/
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -mkdir /Lab05
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Hadoop
ls: `/Hadoop': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Lab05
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ touch test.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ nano text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -put ./text.txt /Lab05/text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Lab05
Found 1 items
-rw-r--r--   1 hadoop supergroup         19 2024-05-13 14:33 /Lab05/text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -cat /Lab05/text.txt
Hello
How are you?
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Lab05
Found 2 items
-rw-r--r--   1 hadoop supergroup         15 2024-05-13 14:40 /Lab05/test.txt
-rw-r--r--   1 hadoop supergroup         19 2024-05-13 14:33 /Lab05/text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -getmerge /Lab05 /text.txt /Lab05 /test.txt ../
Downloads/Merged.txt
getmerge: `/text.txt': No such file or directory
getmerge: `/test.txt': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -getmerge /Lab05/text.txt /Lab05/test.txt ../Do
wnloads/Merged.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -getfacl /Lab05
# file: /Lab05
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -put /home/hadoop/Desktop/Welcome.txt /abc/WC.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -copyFromLocal /home/hadoop/Desktop/Welcome.txt /abc/WC.txt
copyFromLocal: `/abc/WC.txt': File exists
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -get /abc/WC.txt /home/hadoop/Downloads/WWC.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -getmerge /abc/ /home/hadoop/Desktop/Merge.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -getfacl /abc/
# file: /abc
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x

hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -cat /abc/WC.txt
hello world
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -mv /abc /FFF
hadoop fs -ls /FFF
Found 3 items
-rw-r--r--   1 hadoop supergroup         12 2025-04-15 14:53 /FFF/WC.txt
-rw-r--r--   1 hadoop supergroup         12 2024-05-14 14:35 /FFF/file.txt
-rw-r--r--   1 hadoop supergroup         12 2024-05-14 14:38 /FFF/file_cp_local.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -cp /CSE/ /LLL
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -copyToLocal /Lab05/text.txt ../Documents
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -copyToLocal /Lab05/test.txt ../Documents
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -cat /Lab05/text.txt
Hello
How are you?
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -mv /Lab05 /test_Lab05
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -ls /test_Lab05
Found 2 items
-rw-r--r--   1 hadoop supergroup         15 2024-05-13 14:40 /test_Lab05/test.txt
-rw-r--r--   1 hadoop supergroup         19 2024-05-13 14:33 /test_Lab05/text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -cp /test_Lab05/ /Lab05
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -ls /Lab05
Found 2 items
-rw-r--r--   1 hadoop supergroup         15 2024-05-13 14:51 /Lab05/test.txt
-rw-r--r--   1 hadoop supergroup         19 2024-05-13 14:51 /Lab05/text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -ls /test_Lab05
Found 2 items
-rw-r--r--   1 hadoop supergroup         15 2024-05-13 14:40 /test_Lab05/test.txt
-rw-r--r--   1 hadoop supergroup         19 2024-05-13 14:33 /test_Lab05/text.txt
```

LABORATORY PROGRAM – 5

Implement Wordcount program on Hadoop framework

CODE, COMMAND WITH OUTPUT

Driver Code

```
// Importing libraries
import java.io.IOException;


import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool {

    public int run(String[] args) throws IOException {
        if (args.length < 2) {
            System.out.println("Please give valid inputs");
            return -1;
        }

        JobConf conf = new JobConf(WCDriver.class);
        conf.setJobName("WordCount");

        FileInputFormat.setInputPaths(conf, new Path(args[0]));
        FileOutputFormat.setOutputPath(conf, new Path(args[1]));

        conf.setMapperClass(WCMapper.class);
        conf.setReducerClass(WCReducer.class);

        conf.setMapOutputKeyClass(Text.class);
        conf.setMapOutputValueClass(IntWritable.class);

        conf.setOutputKeyClass(Text.class);
        conf.setOutputValueClass(IntWritable.class);

        JobClient.runJob(conf);
        return 0;
    }

    // Main Method
    public static void main(String[] args) throws Exception {
        int exitCode = ToolRunner.run(new WCDriver(), args);
        System.out.println("Job Exit Code: " + exitCode);
    }
}
```

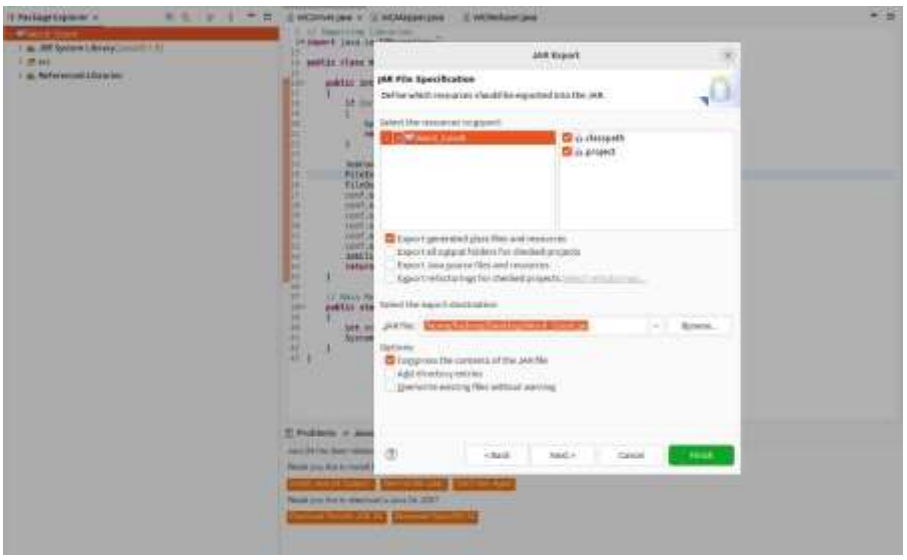
Mapper Code

```
// Importing libraries
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;
```

```
public class WCMapper extends MapReduceBase implements Mapper<LongWritable, Text, Text, IntWritable> {

    // Map function
    public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output, Reporter reporter)
            throws IOException {
        String line = value.toString();

        // Splitting the line on whitespace
        for (String word : line.split("\\s+")) {
            if (word.length() > 0) {
                output.collect(new Text(word), new IntWritable(1));
            }
        }
    }
}
```

Reducer Code

```
// Importing libraries
import java.io.IOException;
import java.util.Iterator;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import  org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;

public class WCReducer extends MapReduceBase implements Reducer<Text, IntWritable, Text, IntWritable> {

    // Reduce function
    public void reduce(Text key, Iterator<IntWritable> values,
                    OutputCollector<Text, IntWritable> output,
                    Reporter reporter) throws IOException {
        int count = 0;

        // Counting the frequency of each word
        while (values.hasNext()) {
            count += values.next().get();
        }

        output.collect(key, new IntWritable(count));
    }
}
```



```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ cd ./Desktop/
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -mkdir /Lab06
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Lab06
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ jps
7360 DataNode
7928 ResourceManager
8681 Jps
7178 NameNode
8091 NodeManager
7644 SecondaryNameNode
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ cd ..
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ cd ./Desktop/
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ nano file1.txt
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -copyFromLocal -f /home/hadoop/Desktop/file1.txt /rgs/test.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop jar /home/hadoop/Desktop/WordCount.jar wordcount.WordCount /rgs/test.txt /output
JAR does not exist or is not a normal file: /home/hadoop/Desktop/WordCount.jar
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop jar /home/hadoop/Desktop/Word_Count.jar wordcount.WordCount /rgs/test.txt /output
Exception in thread "main" java.lang.ClassNotFoundException: wordcount.WordCount
        at java.base/java.net.URLClassLoader.findClass(URLClassLoader.java:476)
        at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:594)
        at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:527)
        at java.base/java.lang.Class.forName0(Native Method)
        at java.base/java.lang.Class.forName(Class.java:398)
        at org.apache.hadoop.util.RunJar.run(RunJar.java:321)
        at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -cat /output/part-00000
are     1
brother 1
family  1
hi      1
how     5
is      4
job     1
sister  1
you     1
your    4
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /output
Found 2 items
-rw-r--r--   1 hadoop supergroup          0 2024-05-21 15:21 /output/_SUCCESS
-rw-r--r--   1 hadoop supergroup         69 2024-05-21 15:21 /output/part-00000
```

LABORATORY PROGRAM – 6

Implement Weather program on Hadoop framework

Questions:

From the following link extract the weather data
https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all
a) Create a MapReduce program to find average temperature for each year from NCDC data set.
b) find the mean max temperature for every month.

CODE, COMMAND WITH OUTPUT – A

Driver Code

```
package temp;


import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class AverageDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Please enter both input and output parameters.");
            System.exit(-1);
        }

        // Creating a configuration and job instance
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Average Calculation");

        job.setJarByClass(AverageDriver.class);

        // Input and output paths
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        // Setting mapper and reducer classes
        job.setMapperClass(AverageMapper.class);
        job.setReducerClass(AverageReducer.class);

        // Output key and value types
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        // Submitting the job and waiting for it to complete
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code

```
package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class AverageMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {

        String line = value.toString();

        // Extract year from fixed position
        String year = line.substring(15, 19);
        int temperature;

        // Determine if there's a '+' sign
        if (line.charAt(87) == '+') {
            temperature = Integer.parseInt(line.substring(88, 92));
        } else {
            temperature = Integer.parseInt(line.substring(87, 92));
        }

        // Quality check character
        String quality = line.substring(92, 93);

        // Only emit if data is valid
        if (temperature != MISSING && quality.matches("[01459]")) {
            context.write(new Text(year), new IntWritable(temperature));
        }
    }
}
```

Reducer Code

```
package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class AverageReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
                    Context context) throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;

        for (IntWritable value : values) {
            sumTemp += value.get();
            count++;
        }

        if (count > 0) {
            int average = sumTemp / count;
            context.write(key, new IntWritable(average));
```

```
        }
    }
}
```

Name	Size	Type	Modified
META-INF	25 bytes	Folder	
.classpath	2.2 kB	unknown	06 May 2025, 14:40
.project	377 bytes	unknown	06 May 2025, 14:34
AverageDriver.class	1.6 kB	Java class	06 May 2025, 14:42
AverageMapper.class	2.4 kB	Java class	06 May 2025, 14:42
AverageReducer.class	2.3 kB	Java class	06 May 2025, 14:42

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ jps
7056 DataNode
7332 SecondaryNameNode
7638 ResourceManager
8231 Jps
5883 org.eclipse.equinox.launcher_1.6.1000.v20250227-1734.jar
7804 NodeManager
6877 NameNode
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /\
> ^C
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /
Found 4 items
drwxr-xr-x   - hadoop supergroup          0 2025-04-15 15:00 /FFF
drwxr-xr-x   - hadoop supergroup          0 2025-04-15 15:34 /LLL
drwxr-xr-x   - hadoop supergroup          0 2024-05-13 14:46 /file
drwxr-xr-x   - hadoop supergroup          0 2024-05-13 15:18 /newDataFlair
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /weather
ls: `/weather': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mkdir /weather
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -copyFromLocal /home/hadoop/Desktop/1901.txt /weather/test.txt
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop jar /home/hadoop/Desktop/AverageTemperature.jar AverageDriver /weather/test.txt /weather/output
2025-05-06 14:59:23,239 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2025-05-06 14:59:23,279 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2025-05-06 14:59:23,279 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-05-06 14:59:23,340 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2025-05-06 14:59:23,393 INFO input.FileInputFormat: Total input files to process : 1
2025-05-06 14:59:23,422 INFO mapreduce.JobSubmitter: number of splits:1
2025-05-06 14:59:23,487 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local91822013_0001
2025-05-06 14:59:23,488 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-05-06 14:59:23,560 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2025-05-06 14:59:23,560 INFO mapreduce.Job: Running job: job_local91822013_0001
2025-05-06 14:59:23,561 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2025-05-06 14:59:23,564 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2025-05-06 14:59:23,565 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-05-06 14:59:23,565 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-05-06 14:59:23,566 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2025-05-06 14:59:23,601 INFO mapred.LocalJobRunner: Waiting for map tasks
2025-05-06 14:59:23,603 INFO mapred.LocalJobRunner: Starting task: attempt_local91822013_0001_m_000000_0
2025-05-06 14:59:23,615 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2025-05-06 14:59:23,616 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-05-06 14:59:23,616 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-05-06 14:59:23,622 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2025-05-06 14:59:23,624 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/weather/test.txt:0+888100
2025-05-06 14:59:23,658 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2025-05-06 14:59:23,658 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2025-05-06 14:59:23,658 INFO mapred.MapTask: soft limit at 83886080
2025-05-06 14:59:23,658 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2025-05-06 14:59:23,658 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2025-05-06 14:59:23,660 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
```

```
2025-05-06 14:59:24,581 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=153118
                FILE: Number of bytes written=1493804
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=1776380
                HDFS: Number of bytes written=8
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=6565
                Map output records=6564
                Map output bytes=59076
                Map output materialized bytes=72210
                Input split bytes=103
                Combine input records=0
                Combine output records=0
                Reduce input groups=1
                Reduce shuffle bytes=72210
                Reduce input records=6564
                Reduce output records=1
                Spilled Records=13128
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=1266679808
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=888190
        File Output Format Counters
                Bytes Written=8
```

```
                Bytes Written=8
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /weather
Found 2 items
drwxr-xr-x   - hadoop supergroup          0 2025-05-06 14:59 /weather/output
-rw-r--r--   1 hadoop supergroup     888190 2025-05-06 14:50 /weather/test.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /weather/output
Found 2 items
-rw-r--r--   1 hadoop supergroup          0 2025-05-06 14:59 /weather/output/_SUCCESS
-rw-r--r--   1 hadoop supergroup          8 2025-05-06 14:59 /weather/output/part-r-00000
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cat /weather/output/part-r-00000
1901    46
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

CODE, COMMAND WITH OUTPUT – B

Driver Code

```
package meanmax;


import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class MeanMaxDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Please enter both input and output parameters.");
            System.exit(-1);
        }

        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Mean and Max Temperature");

        job.setJarByClass(MeanMaxDriver.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        job.setMapperClass(MeanMaxMapper.class);
        job.setReducerClass(MeanMaxReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code

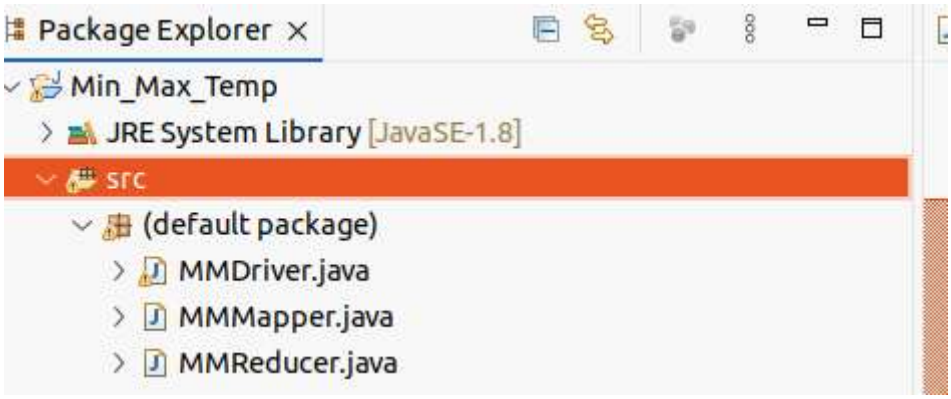
```
package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Mapper;

public class MeanMaxMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {

        String line = value.toString();

        // Extract month from positions 19-20
        String month = line.substring(19, 21);
```

```
int temperature;
```

```
        // Extract temperature considering optional '+'
        if (line.charAt(87) == '+') {
            temperature = Integer.parseInt(line.substring(88, 92));
        } else {
            temperature = Integer.parseInt(line.substring(87, 92));
        }

        // Quality check
        String quality = line.substring(92, 93);

        if (temperature != MISSING && quality.matches("[01459]")) {
            context.write(new Text(month), new IntWritable(temperature));
        }
    }
}
```

Reducer Code

```
package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class MeanMaxReducer extends Reducer<Text, IntWritable, Text, Text> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
                    Context context) throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;
        int maxTemp = Integer.MIN_VALUE;

        for (IntWritable value : values) {
            int temp = value.get();
            sumTemp += temp;
            count++;

            if (temp > maxTemp) {
                maxTemp = temp;
            }
        }

        if (count > 0) {
            int avgTemp = sumTemp / count;
            String result = "mean=" + avgTemp + " max=" + maxTemp;
            context.write(key, new Text(result));
        }
    }
}
```



```
hadoop@mescerce-HP-Elite-Tower-800-G9-Desktop-PC: $ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
localhost: namenode is running as process 5470.  Stop it first and ensure /tmp/hadoop-hadoop-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 5644.  Stop it first and ensure /tmp/hadoop-hadoop-datanode.pid file is empty before retry.
Starting secondary namenodes [mescerce-HP-Elite-Tower-800-G9-Desktop-PC]
mescerce-HP-Elite-Tower-800-G9-Desktop-PC: secondarynamenode is running as process 5931.  Stop it first and ensure /tmp/hadoop-hadoop-secondarynamenode.pid file is empty before retry.
Starting resourcemanager
resourcemanager is running as process 6214.  Stop it first and ensure /tmp/hadoop-hadoop-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 6375.  Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file is empty before retry.
hadoop@mescerce-HP-Elite-Tower-800-G9-Desktop-PC: $ hdfs dfs -copyFromLocal /home/hadoop/Desktop/1901 /rgs/temp
copyFromLocal: `/rgs/temp': File exists
hadoop@mescerce-HP-Elite-Tower-800-G9-Desktop-PC: $ hdfs dfs -mkdir -p /rgs
hadoop@mescerce-HP-Elite-Tower-800-G9-Desktop-PC: $ hdfs dfs -copyFromLocal /home/hadoop/Desktop/1901 /rgs/1901
hadoop@mescerce-HP-Elite-Tower-800-G9-Desktop-PC: $ hadoop jar /home/hduser/Downloads/Mean_Max_Temp.jar MMDriver /rgs/avtemp.txt /out8
JAR does not exist or is not a normal file: /home/hduser/Downloads/Mean_Max_Temp.jar
hadoop@mescerce-HP-Elite-Tower-800-G9-Desktop-PC: $ hadoop jar /home/hduser/Downloads/Min_Max_Temp.jar MMDriver /rgs/avtemp.txt /out8
JAR does not exist or is not a normal file: /home/hduser/Downloads/Min_Max_Temp.jar
hadoop@mescerce-HP-Elite-Tower-800-G9-Desktop-PC: $ hadoop jar /home/hadoop/Desktop/Min_Max_Temp.jar MMDriver /rgs/avtemp.txt /out8
2025-05-06 15:23:85,439 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2025-05-06 15:23:85,471 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2025-05-06 15:23:85,471 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-05-06 15:23:85,531 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2025-05-06 15:23:85,571 INFO mapreduce.JobSubmitter: Cleaning up the staging area file:/tmp/hadoop/mapred/staging/hadoop1762065270/.staging/job_local1762065270_0001
Exception in thread "main" org.apache.hadoop.mapreduce.lib.input.InvalidInputException: Input path does not exist: hdfs://localhost:9000/rgs/avtemp.txt
        at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.singleThreadedListStatus(FileInputFormat.java:340)
        at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.listStatus(FileInputFormat.java:279)
        at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.getSplits(FileInputFormat.java:404)
        at org.apache.hadoop.mapreduce.JobSubmitter.writeNewSplits(JobSubmitter.java:310)
        at org.apache.hadoop.mapreduce.JobSubmitter.writeSplits(JobSubmitter.java:327)
        at org.apache.hadoop.mapreduce.JobSubmitter.submitJobInternal(JobSubmitter.java:200)
        at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1670)
        at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1675)
        at java.base/java.security.AccessController.doPrivileged(Native Method)
        at java.base/javax.security.auth.Subject.doAs(Subject.java:423)
        at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1899)
        at org.apache.hadoop.mapreduce.Job.submit(Job.java:1675)
        at org.apache.hadoop.mapreduce.Job.waitForCompletion(Job.java:1696)
        at MMDriver.main(MMDriver.java:40)
        at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
        at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:62)
        at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
        at java.base/java.lang.reflect.Method.invoke(Method.java:566)
        at org.apache.hadoop.util.RunJar.run(RunJar.java:328)
        at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
Caused by: java.io.IOException: Input path does not exist: hdfs://localhost:9000/rgs/avtemp.txt
        at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.singleThreadedListStatus(FileInputFormat.java:310)
        ... 19 more
hadoop@mescerce-HP-Elite-Tower-800-G9-Desktop-PC: $ hdfs dfs -cat /out8/*
cat: `/out8/*': No such file or directory
hadoop@mescerce-HP-Elite-Tower-800-G9-Desktop-PC: $ ^C
hadoop@mescerce-HP-Elite-Tower-800-G9-Desktop-PC: $ hdfs dfs -ls /
```

```
Caused by: java.io.IOException: Input path does not exist: hdfs://localhost:9000/rgs/avtemp.txt
        at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.singleThreadedListStatus(FileInputFormat.java:310)
        ... 19 more
hadoop@mescerce-HP-Elite-Tower-800-G9-Desktop-PC: $ hadoop jar /home/hadoop/Desktop/Min_Max_Temp.jar MMDriver /rgs/1901 /out8
2025-05-06 15:26:34,076 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2025-05-06 15:26:34,910 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2025-05-06 15:26:34,910 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-05-06 15:26:34,976 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2025-05-06 15:26:35,029 INFO input.FileInputFormat: Total input files to process : 1
2025-05-06 15:26:35,081 INFO mapreduce.JobSubmitter: number of splits:1
2025-05-06 15:26:35,148 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1863792110_0001
2025-05-06 15:26:35,148 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-05-06 15:26:35,211 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2025-05-06 15:26:35,216 INFO mapreduce.Job: Running job: job_local1863792110_0001
2025-05-06 15:26:35,216 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2025-05-06 15:26:35,220 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2025-05-06 15:26:35,221 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-05-06 15:26:35,221 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-05-06 15:26:35,221 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2025-05-06 15:26:35,287 INFO mapred.LocalJobRunner: Waiting for map tasks
2025-05-06 15:26:35,288 INFO mapred.LocalJobRunner: Starting task: attempt_local1863792110_0001_m_000000_0
2025-05-06 15:26:35,276 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2025-05-06 15:26:35,278 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-05-06 15:26:35,279 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-05-06 15:26:35,286 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2025-05-06 15:26:35,287 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/rgs/1901:0+600190
2025-05-06 15:26:35,322 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2025-05-06 15:26:35,322 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2025-05-06 15:26:35,322 INFO mapred.MapTask: soft limit at 83886080
2025-05-06 15:26:35,322 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2025-05-06 15:26:35,322 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2025-05-06 15:26:35,325 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2025-05-06 15:26:35,440 INFO mapred.LocalJobRunner:
2025-05-06 15:26:35,441 INFO mapred.MapTask: Starting flush of map output
2025-05-06 15:26:35,441 INFO mapred.MapTask: Spilling map output
2025-05-06 15:26:35,441 INFO mapred.MapTask: bufstart = 0; bufend = 45948; bufvoid = 104857600
2025-05-06 15:26:35,441 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26188144(104752576); length = 26253/6553600
2025-05-06 15:26:35,451 INFO mapred.MapTask: Finished spill 0
2025-05-06 15:26:35,456 INFO mapred.Task: Task:attempt_local1863792110_0001_m_000000_0 is done. And is in the process of committing
2025-05-06 15:26:35,458 INFO mapred.LocalJobRunner: map
2025-05-06 15:26:35,458 INFO mapred.Task: Task 'attempt_local1863792110_0001_m_000000_0' done.
2025-05-06 15:26:35,461 INFO mapred.Task: Final Counters for attempt_local1863792110_0001_m_000000_0: Counters: 23
        File System Counters
                FILE: Number of bytes read=4320
                FILE: Number of bytes written=703893
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=600190
                HDFS: Number of bytes written=0
                HDFS: Number of read operations=1
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=1
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=1
```

```
2025-05-06 15:26:36,233 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=126914
                FILE: Number of bytes written=1466688
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=177638B
                HDFS: Number of bytes written=74
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=6565
                Map output records=6564
                Map output bytes=45948
                Map output materialized bytes=59082
                Input split bytes=95
                Combine input records=0
                Combine output records=0
                Reduce input groups=12
                Reduce shuffle bytes=59082
                Reduce input records=6564
                Reduce output records=12
                Spilled Records=13128
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=1052770304
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=888190
        File Output Format Counters
                Bytes Written=74
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /out8/*
01      4
02      0
03      7
04      44
05      100
06      168
07      219
08      198
09      141
10      100
11      19
12      3
```

LABORATORY PROGRAM – 7

For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words.

CODE, COMMAND WITH OUTPUT

Driver Code (TopNDriver.java)

```
package samples.topn;


import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopNDriver {

    public static void main(String[] args) throws Exception {
        if (args.length != 3) {
            System.err.println("Usage: TopNDriver <in> <temp-out> <final-out>");
            System.exit(2);
        }

        Configuration conf = new Configuration();

        // === Job 1: Word Count ===
        Job wcJob = Job.getInstance(conf, "word count");
        wcJob.setJarByClass(TopNDriver.class);
        wcJob.setMapperClass(WordCountMapper.class);
        wcJob.setCombinerClass(WordCountReducer.class);
        wcJob.setReducerClass(WordCountReducer.class);
        wcJob.setOutputKeyClass(Text.class);
        wcJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(wcJob, new Path(args[0]));
        Path tempDir = new Path(args[1]);
        FileOutputFormat.setOutputPath(wcJob, tempDir);

        if (!wcJob.waitForCompletion(true)) {
            System.exit(1);
        }

        // === Job 2: Top N ===
        Job topJob = Job.getInstance(conf, "top 10 words");
        topJob.setJarByClass(TopNDriver.class);
        topJob.setMapperClass(TopNMapper.class);
        topJob.setReducerClass(TopNReducer.class);
        topJob.setMapOutputKeyClass(IntWritable.class);
        topJob.setMapOutputValueClass(Text.class);
        topJob.setOutputKeyClass(Text.class);
        topJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(topJob, tempDir);
        FileOutputFormat.setOutputPath(topJob, new Path(args[2]));

        System.exit(topJob.waitForCompletion(true) ? 0 : 1);
    }
```

}

Mapper Code (WordCountMapper.java)

```
package samples.topn;

import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class WordCountMapper
    extends Mapper<Object, Text, Text, IntWritable> {

    private final static IntWritable ONE = new IntWritable(1);
    private Text word = new Text();
    // characters to normalize into spaces
    private String tokens = "[_|$#<>\\^=\\[\\]\\*/\\\\,;,.\\-:()?!\"]";

    @Override
    protected void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        // clean & tokenize
        String clean = value.toString()
                        .toLowerCase()
                        .replaceAll(tokens, " ");
        StringTokenizer itr = new StringTokenizer(clean);
        while (itr.hasMoreTokens()) {
            word.set(itr.nextToken().trim());
            context.write(word, ONE);
        }
    }
}
```

Mapper Code (TopNMapper.java)

```
package samples.topn;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class TopNMapper
    extends Mapper<Object, Text, IntWritable, Text> {

    private IntWritable count = new IntWritable();
    private Text word = new Text();

    @Override
    protected void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        // input line: word \t count
        String[] parts = value.toString().split("\\t");
        if (parts.length == 2) {
            word.set(parts[0]);
            count.set(Integer.parseInt(parts[1]));
            // emit count → word, so Hadoop sorts by count
            context.write(count, word);
        }
    }
}
```

Reducer Code (WordCountReducer.java)

```
package samples.topn;

import java.io.IOException;
```

```
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class WordCountReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

        int sum = 0;
        for (IntWritable val : values) {
            sum += val.get();
        }
        context.write(key, new IntWritable(sum));
    }
}
```

Reducer Code (TopNReducer.java)

```
package samples.topn;

import java.io.IOException;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.Map;
import java.util.TreeMap;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class TopNReducer
    extends Reducer<IntWritable, Text, Text, IntWritable> {

    // TreeMap with descending order of keys (counts)
    private TreeMap<Integer, List<String>> countMap =
        new TreeMap<>(Collections.reverseOrder());

    @Override
    protected void reduce(IntWritable key, Iterable<Text> values, Context context)
        throws IOException, InterruptedException {

        int cnt = key.get();
        List<String> words = countMap.getOrDefault(cnt, new ArrayList<>());
        for (Text w : values) {
            words.add(w.toString());
        }
        countMap.put(cnt, words);
    }

    @Override
    protected void cleanup(Context context)
        throws IOException, InterruptedException {

        // collect top 10 word→count pairs
        List<WordCount> topList = new ArrayList<>();
        int seen = 0;
        for (Map.Entry<Integer, List<String>> entry : countMap.entrySet()) {
            int cnt = entry.getKey();
            for (String w : entry.getValue()) {
                topList.add(new WordCount(w, cnt));
                seen++;
                if (seen == 10) break;
            }
            if (seen == 10) break;
```

```
        }

        // sort these 10 entries alphabetically by word
        Collections.sort(topList, (a, b) -> a.word.compareTo(b.word));

        // emit final top 10 in alphabetical order
        for (WordCount wc : topList) {
            context.write(new Text(wc.word), new IntWritable(wc.count));
        }
    }

    // helper class
    private static class WordCount {
        String word;
        int count;
        WordCount(String w, int c) { word = w; count = c; }
    }
}
```

```
C:\hadoop-3.3.0\sbin>jps
11072 DataNode
20528 Jps
5620 ResourceManager
15532 NodeManager
6140 NameNode

C:\hadoop-3.3.0\sbin>hdfs dfs -mkdir /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -ls /
Found 1 items
drwxr-xr-x   - Anusree supergroup          0 2021-05-08 19:46 /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -copyFromLocal C:\input.txt /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -ls /input_dir
Found 1 items
-rw-r--r--   1 Anusree supergroup         36 2021-05-08 19:48 /input_dir/input.txt

C:\hadoop-3.3.0\sbin>hdfs dfs -cat /input_dir/input.txt
hello
world
hello
hadoop
bye
```

```
C:\hadoop-3.3.0\sbin>hadoop jar C:\sort.jar samples.topn.TopN /input_dir/input.txt /output_dir
2021-05-08 19:54:54,582 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2021-05-08 19:54:55,291 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anusree/.staging/job_1620483374279_0001
2021-05-08 19:54:55,821 INFO input.FileInputFormat: Total input files to process : 1
2021-05-08 19:54:56,261 INFO mapreduce.JobSubmitter: number of splits:1
2021-05-08 19:54:56,552 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1620483374279_0001
2021-05-08 19:54:56,552 INFO mapreduce.JobSubmitter: Executing with tokens: []
2021-05-08 19:54:56,843 INFO conf.Configuration: resource-types.xml not found
2021-05-08 19:54:56,843 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2021-05-08 19:54:57,387 INFO impl.YarnClientImpl: Submitted application application_1620483374279_0001
2021-05-08 19:54:57,507 INFO mapreduce.Job: The url to track the job: http://LAPTOP-JG329ESD:8088/proxy/application_1620483374279_0001/
2021-05-08 19:54:57,508 INFO mapreduce.Job: Running job: job_1620483374279_0001
2021-05-08 19:55:13,792 INFO mapreduce.Job: Job job_1620483374279_0001 running in uber mode : false
2021-05-08 19:55:13,794 INFO mapreduce.Job:  map 0% reduce 0%
2021-05-08 19:55:20,020 INFO mapreduce.Job:  map 100% reduce 0%
2021-05-08 19:55:27,116 INFO mapreduce.Job:  map 100% reduce 100%
2021-05-08 19:55:33,199 INFO mapreduce.Job: Job job_1620483374279_0001 completed successfully
2021-05-08 19:55:33,334 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=65
                FILE: Number of bytes written=530397
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=142
                HDFS: Number of bytes written=31
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
```

```
C:\hadoop-3.3.0\sbin>hdfs dfs -cat /output_dir/*
hello   2
hadoop  1
world   1
bye     1

C:\hadoop-3.3.0\sbin>
```

LABORATORY PROGRAM – 8

Write a Scala program to print numbers from 1 to 100 using for loop.

CODE, COMMAND WITH OUTPUT

```
object PrintNumbers {
  def main(args: Array[String]): Unit = {

    for(i <- 1 to 100) {
      println(i)
    }

  }
}
```

LABORATORY PROGRAM – 9

Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.

CODE, COMMAND WITH OUTPUT

```
val textFile = sc.textFile("/home/bhoom/Desktop/wc.txt")

val counts = textFile.flatMap(line => line.split(" ")).map(word => (word,1)).reduceByKey(_
+ _)

import scala.collection.immutable.ListMap

 val sorted=ListMap(counts.collect.sortWith(_._2 > _._2):_*)// sort in descending order
based on values

println(sorted)

for((k,v)<-sorted) {

if(v>4) {

print(k+",")

print(v)

println()

}

}
```

```
scala> val rdd = spark.sparkContext.textFile("file:/home/bmscecse/Desktop/scala")
rdd: org.apache.spark.rdd.RDD[String] = file:/home/bmscecse/Desktop/scala MapPartitionsRDD[1] at textFile at <console>:23

scala> val counts = rdd.flatMap(_.split("\\s+")).map(word => (word.toLowerCase, 1)).reduceByKey(_ + _).filter(_._2 > 4)
counts: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[5] at filter at <console>:25

scala> counts.collect().foreach{ case (word, count) => println(s"$word $count") }
spark 6

scala>
```

LABORATORY PROGRAM – 10

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).

CODE, COMMAND WITH OUTPUT

```
# Install NLTK and download required data (run once)
!pip install nltk

import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

from pyspark.sql import SparkSession
from pyspark.sql.functions import col, lower, regexp_replace, split, explode, udf
from pyspark.sql.types import ArrayType, StringType
from pyspark.ml.feature import StopWordsRemover
from nltk.stem import WordNetLemmatizer

# Initialize SparkSession
spark = SparkSession.builder.appName("TextProcessing").getOrCreate()

# Define your input lines
lines = [
    "Hello, I hate you.",
    "I hate that I love you.",
    "Don't want to, but I can't put",
    "nobody else above you."
]

# Create DataFrame from lines
df = spark.createDataFrame(lines, "string").toDF("value")

# Step 1: Lowercase and remove punctuation
df_clean = df.select(regexp_replace(lower(col("value")), "[^a-zA-Z\\s]", "").alias("cleaned"))

# Step 2: Tokenize the cleaned text
df_tokens = df_clean.select(split(col("cleaned"), "\\s+").alias("tokens"))

# Step 3: Remove stop words
remover = StopWordsRemover(inputCol="tokens", outputCol="filtered")
df_filtered = remover.transform(df_tokens)

# Step 4: Lemmatization using NLTK WordNetLemmatizer with UDF
lemmatizer = WordNetLemmatizer()

def lemmatize_words(words):
    return [lemmatizer.lemmatize(word) for word in words]

lemmatize_udf = udf(lemmatize_words, ArrayType(StringType()))

df_lemmatized = df_filtered.withColumn("lemmatized", lemmatize_udf(col("filtered")))
```

# Step 5: Explode the lemmatized words and show results
df_lemmatized.select(explode(col("lemmatized")).alias("word")).show(truncate=False)

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.11/dist-packages (from nltk) (8.2.0)
Requirement already satisfied: joblib in /usr/local/lib/python3.11/dist-packages (from nltk) (1.5.0)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.11/dist-packages (from nltk) (2024.11.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
+------+
|word  |
+------+
|hello |
|hate  |
|hate  |
|love  |
|dont  |
|want  |
|cant  |
|put   |
|nobody|
|else  |
+------+
```